

Name
Date
Course

Implicit Differentiation (Homework)

Find y' using implicit differentiation.

① $y^5 + 1 = x^2$

② $-6x^3 + y = 3y^2$

③ $4y^5 - 2x - 10 = \sin y - 10x^5$

$$\textcircled{4} \tan x + 5x^2 = 10y^8 + \tan y$$

$$\textcircled{5} 9^x + 1 - y^2 = e^{5x}$$

$$\textcircled{6} e^y + 10^{\sin x} = e^x$$

$$\textcircled{7} 1 + e^{x^2+2x+1} + 5^y = x - y \quad \textcircled{8} 2 - \ln y = y^3 + \log_3 x$$

$$\textcircled{9} 5 \ln x + \log_4 y = \ln(x^2) + \log_8(-5x+2)$$

$$\textcircled{10} \quad 4 + 2x = y \cos x - 3y^6$$

$$\textcircled{11} \quad -5x^3y^3 = 1 + y$$

$$\textcircled{12} \frac{x}{y^5 - 2y} = 4 - 6x^{10}$$

$$\textcircled{13} 2 + \sec y = \frac{x-2}{y-2}$$

Homework Answers

$$\textcircled{1} y' = \frac{2x}{5y^4}$$

$$\textcircled{8} y' = \frac{-\frac{1}{x \ln 3}}{3y^2 + \frac{1}{y}}$$

$$\textcircled{2} y' = -\frac{18x^2}{6y-1} \text{ or } \frac{18x^2}{1-6y}$$

$$\text{or } -\frac{y}{3y^3 x \ln 3 + x \ln 3}$$

$$\textcircled{3} y' = \frac{2-50x^4}{20y^4 - \cos y}$$

$$\textcircled{9} y' = \frac{\frac{2}{x} - \frac{5}{x} - \frac{5}{(-5x+2) \ln 8}}{\frac{1}{y \ln 4}}$$

$$\textcircled{4} y' = \frac{\sec^2 x + 10x}{80y^7 + \sec^2 y}$$

$$y' = \frac{-\frac{3}{x} - \frac{5}{(-5x+2) \ln 8}}{\frac{1}{y \ln 4}}$$

$$\textcircled{5} y' = \frac{9^x \ln 9 - 5e^{5x}}{2y}$$

or

$$\textcircled{6} y' = \frac{e^x - 10^{\sin x} \ln(10) \cos x}{e^y}$$

$$-\frac{3y \ln 4}{x} - \frac{5y \ln 4}{(-5x+2) \ln 8}$$

$$\textcircled{7} y' = \frac{1 - (2x+2)e^{x^2+2x+1}}{5^y \ln 5 + 1}$$

$$\textcircled{10} y' = \frac{2 + y \sin x}{\cos x - 18y^5}$$

$$\text{or } \frac{1 - 2e^{x^2+2x+1}(x+1)}{5^y \ln 5 + 1}$$

$$\textcircled{11} y' = -\frac{15x^2 y^3}{1 + 15x^3 y^2}$$

$$\textcircled{12} y' = \frac{60x^9(y^5-2y)^2 + y^5-2y}{5xy^4-2x}$$

$$\textcircled{13} y' = \frac{y-2}{(y-2)^2 \sec y + \tan y + x-2}$$